

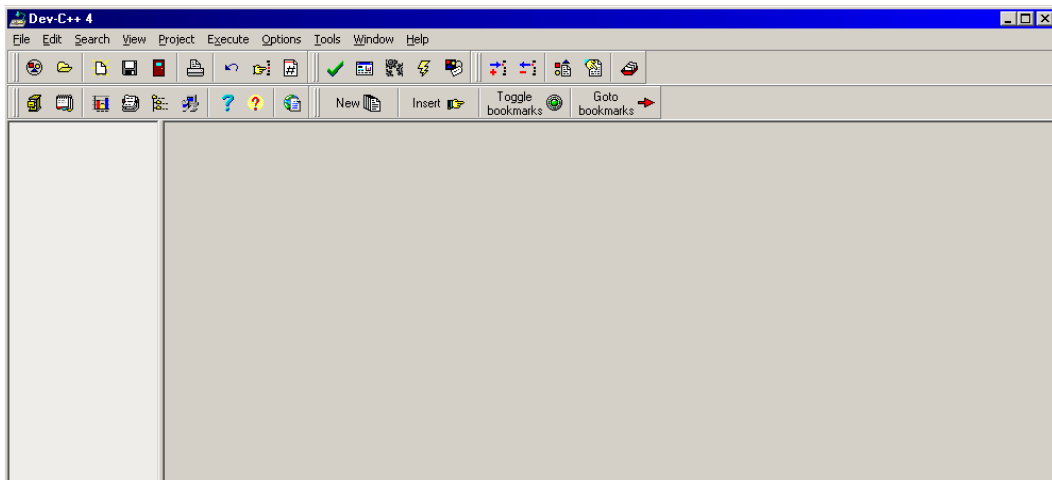
The University of the South Pacific

School of Computing, Information & Mathematical Sciences

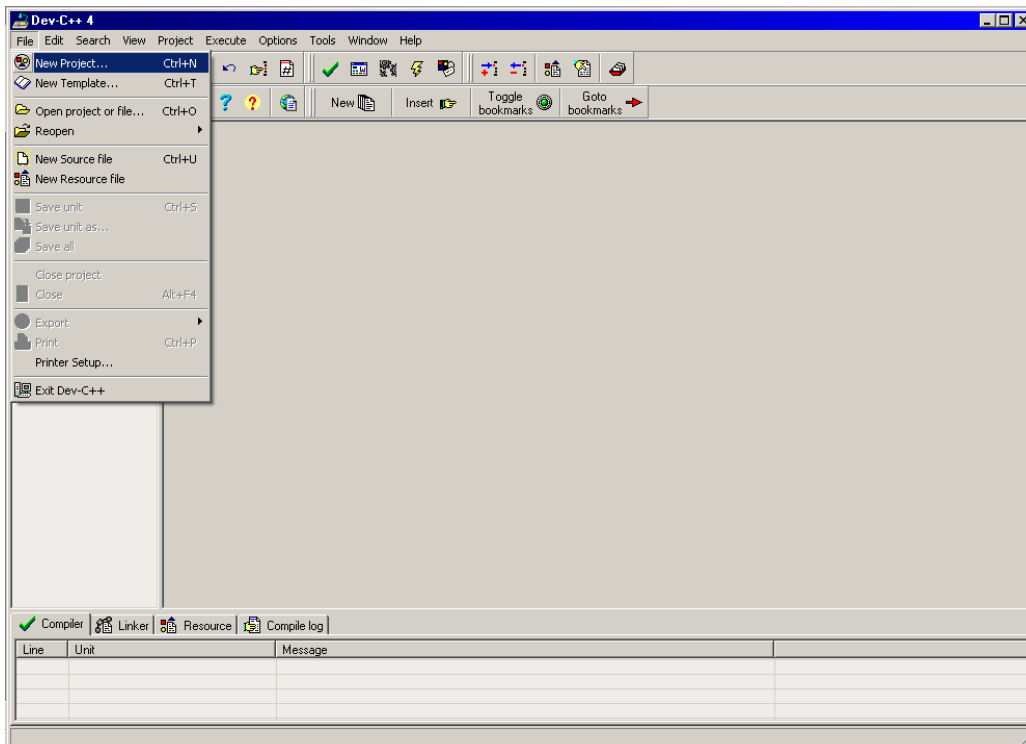
CS 112 Data Structures and Algorithms

Instructions to create project in Dev C++ 4.0 without preprocessor directives

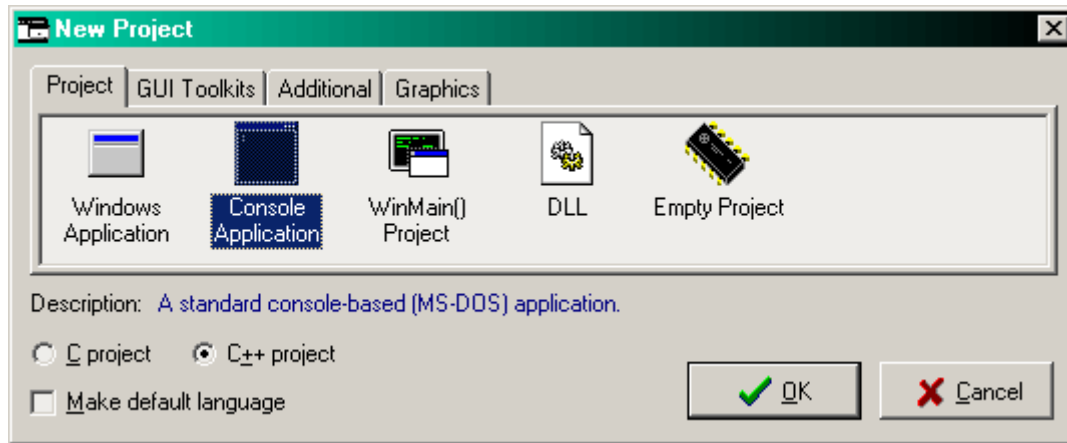
Open Dev C++ by double clicking its icon in desktop or select it from program menu.



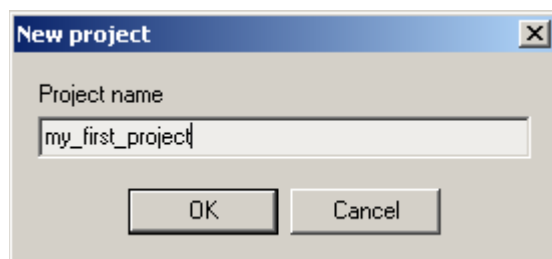
To create C++ project, click *New Project* on the *File* menu as shown below:



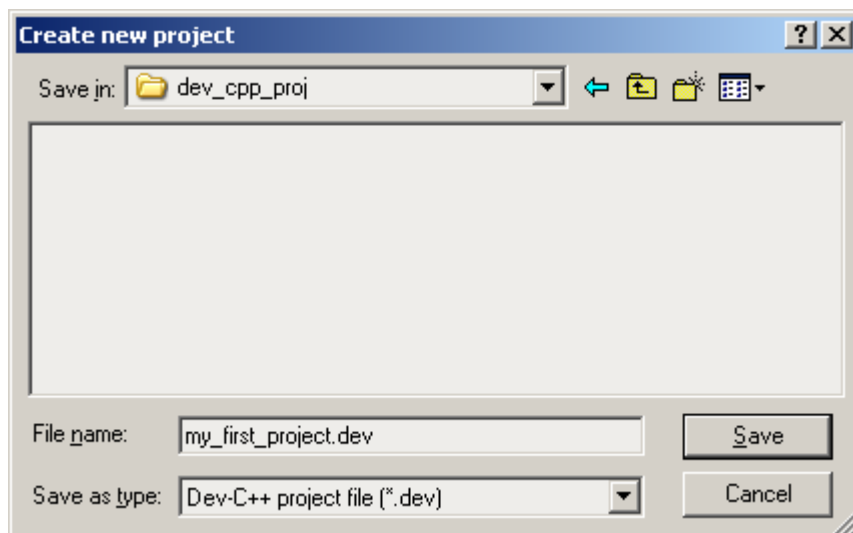
You will see the following pop up dialog box. Choose *Console Application* from the *Project* tab then click OK.



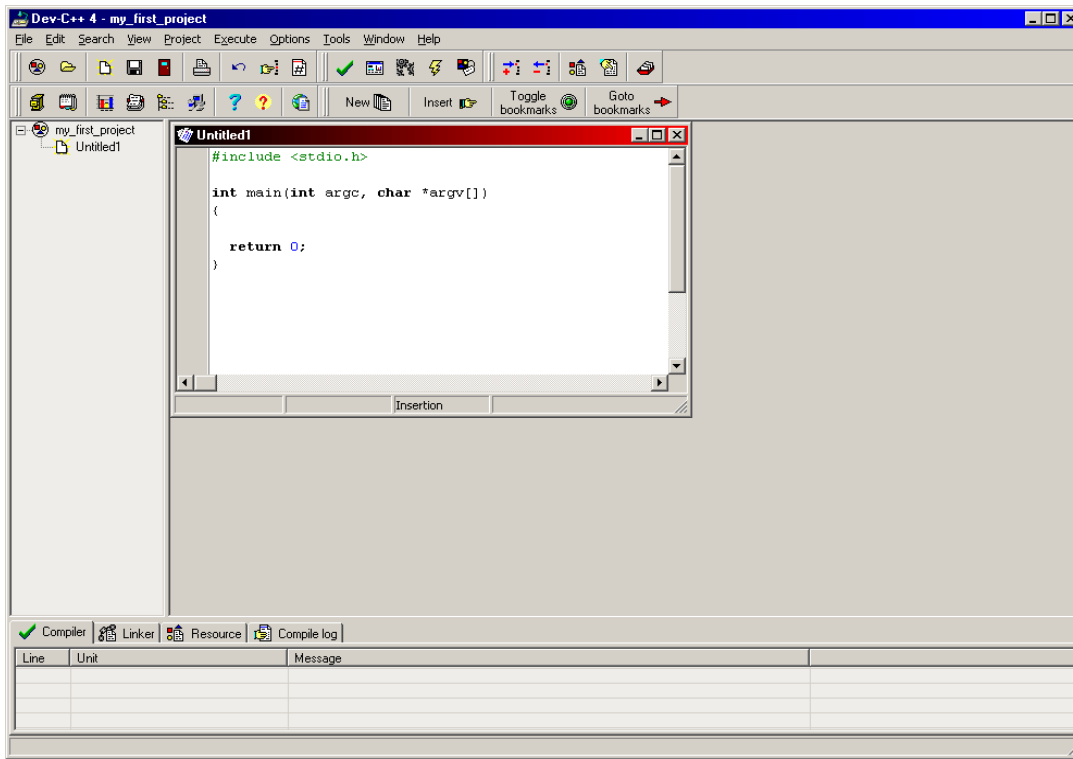
You will be prompted to specify the name of your project. Here `my_first_project` name is used. Click OK after writing project name.



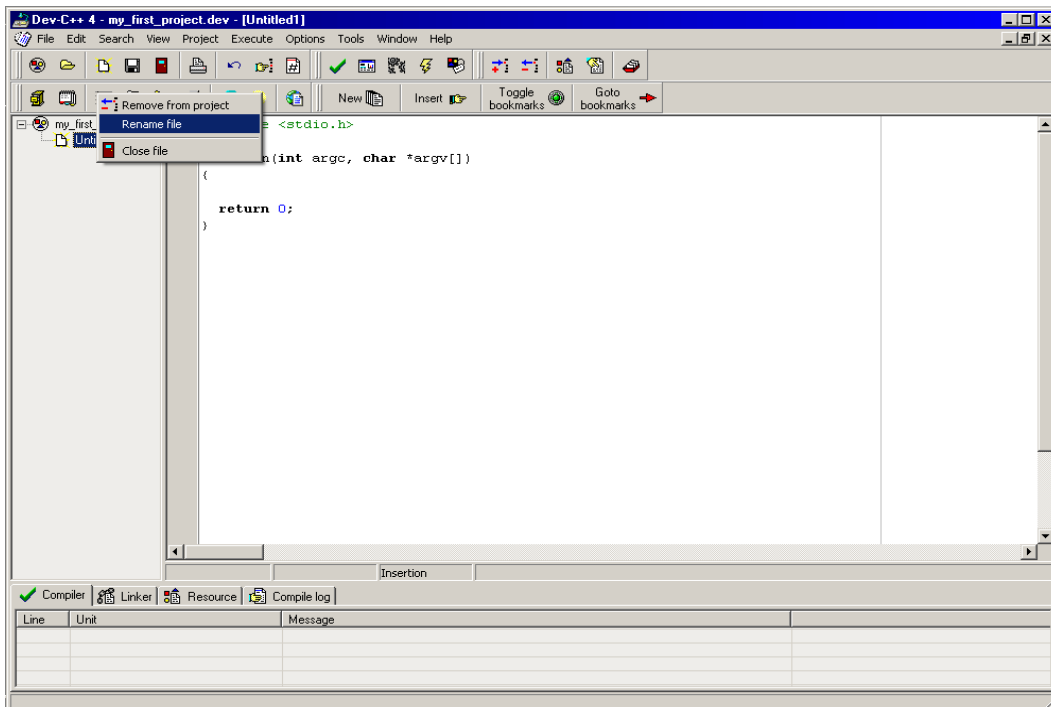
Now save the project in your flash drive or desktop or wherever you like to save.



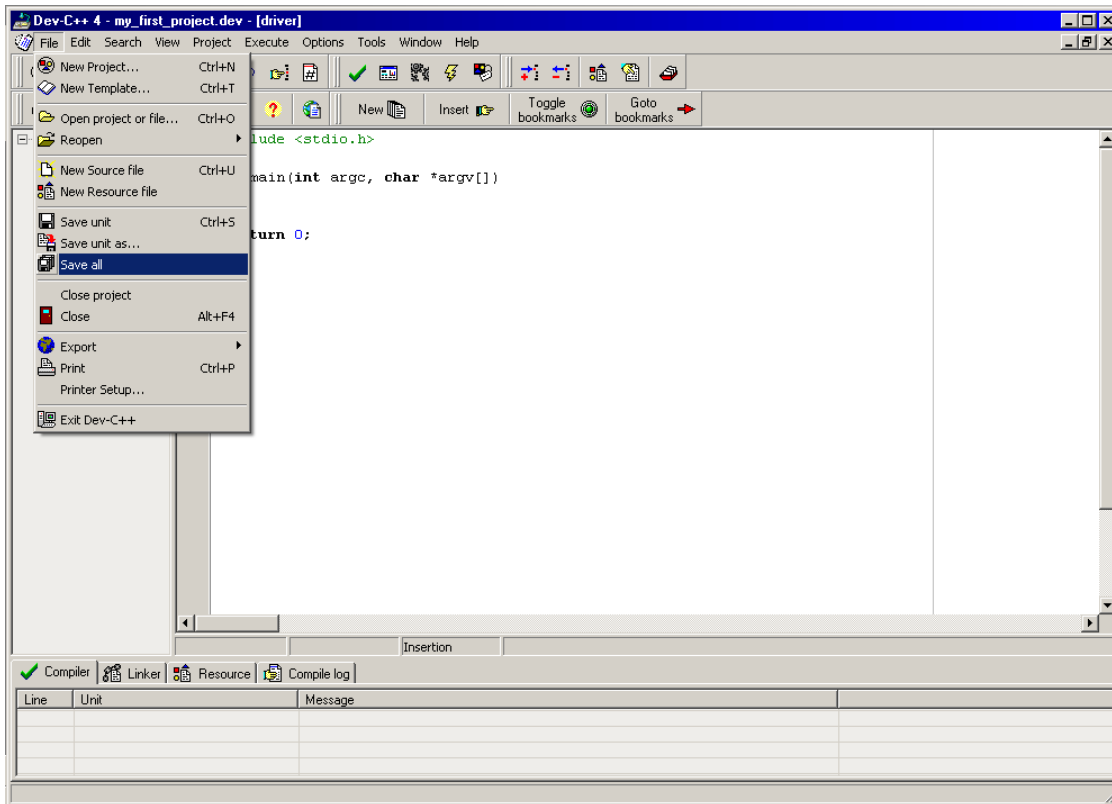
You will see the following window. See the project pane in left hand side. Here you can view all the files of your project. By default a new Untitled1 file is created.



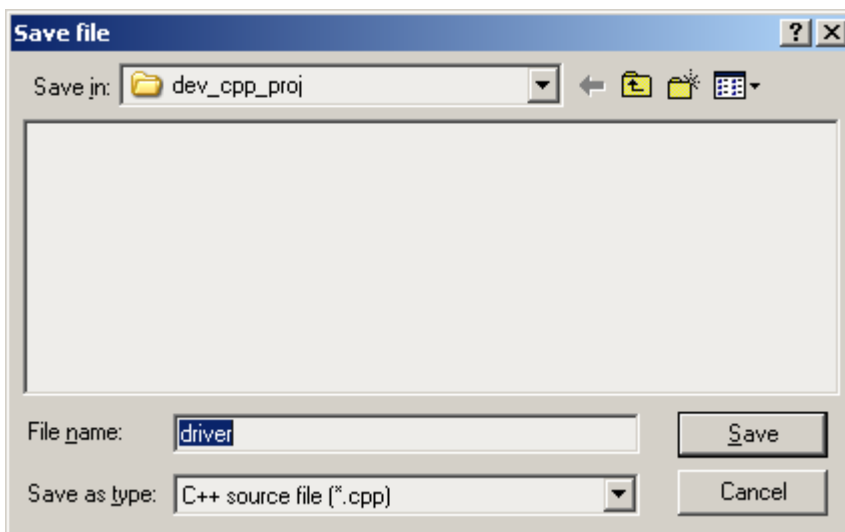
You can save this file by clicking save button or rename it with different file name. Right-click Untitled1 file in left pane then select rename option as shown below:



Type driver in pop-up input box. Now select “Save all” option from File menu as shown below:

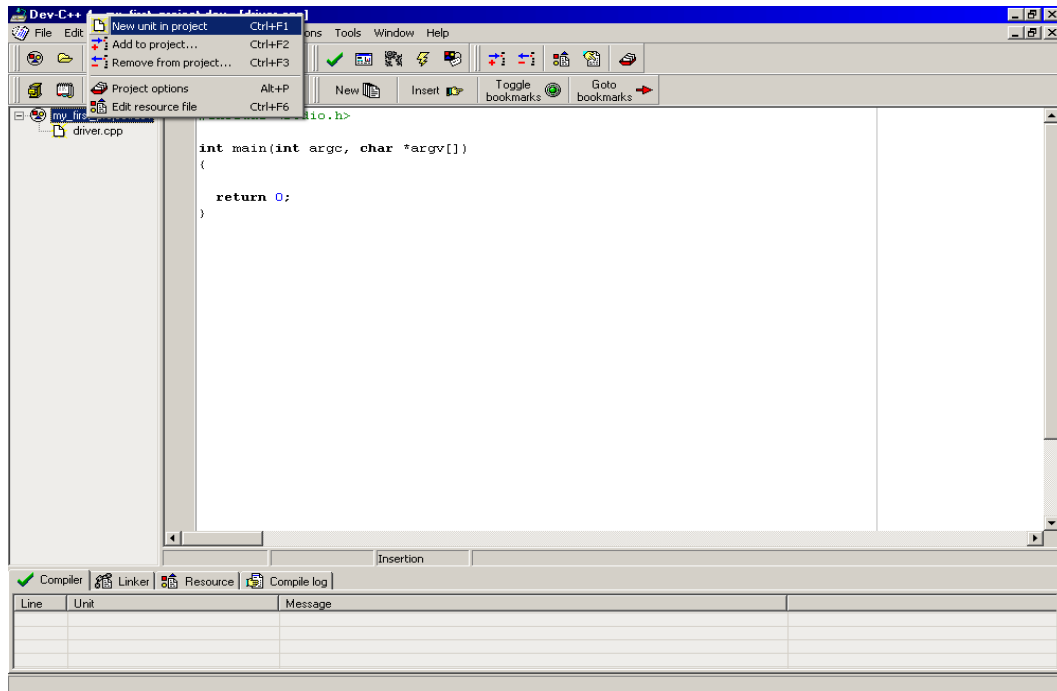


You will be prompted to save the newly created/rename file. Save the file with same name i.e. “driver” in the same project folder.

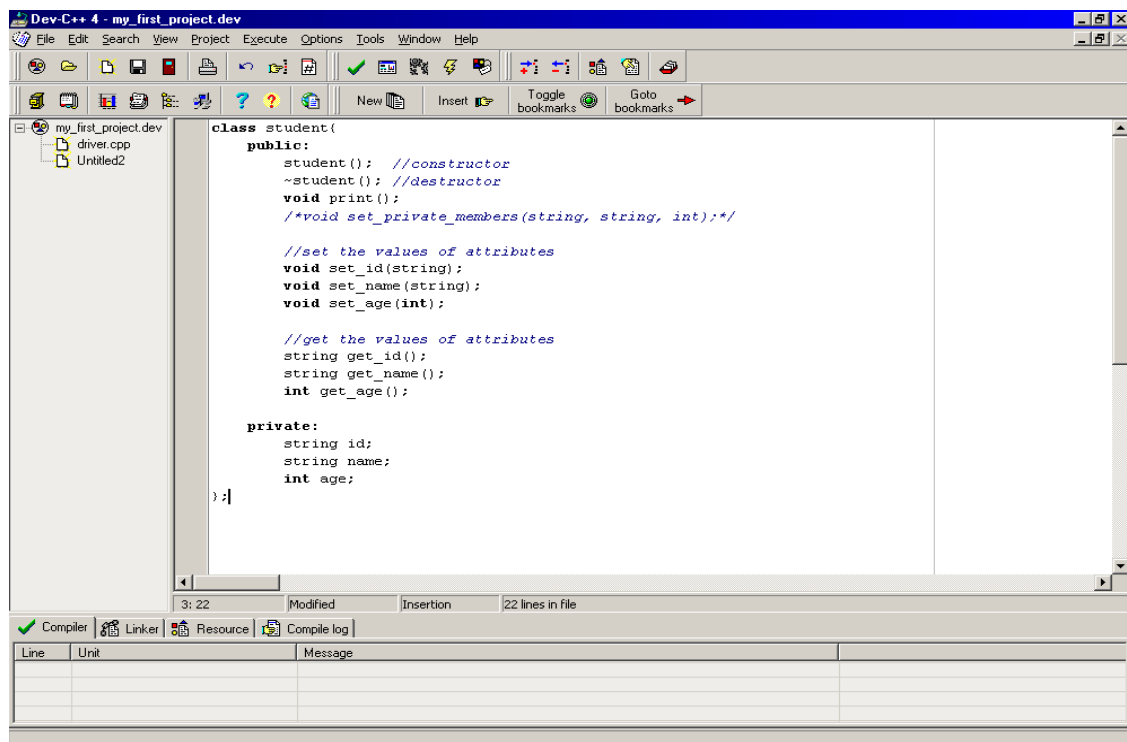


This driver.cpp looks bit different from your traditional C++ program. Don't worry we will deal with it later.

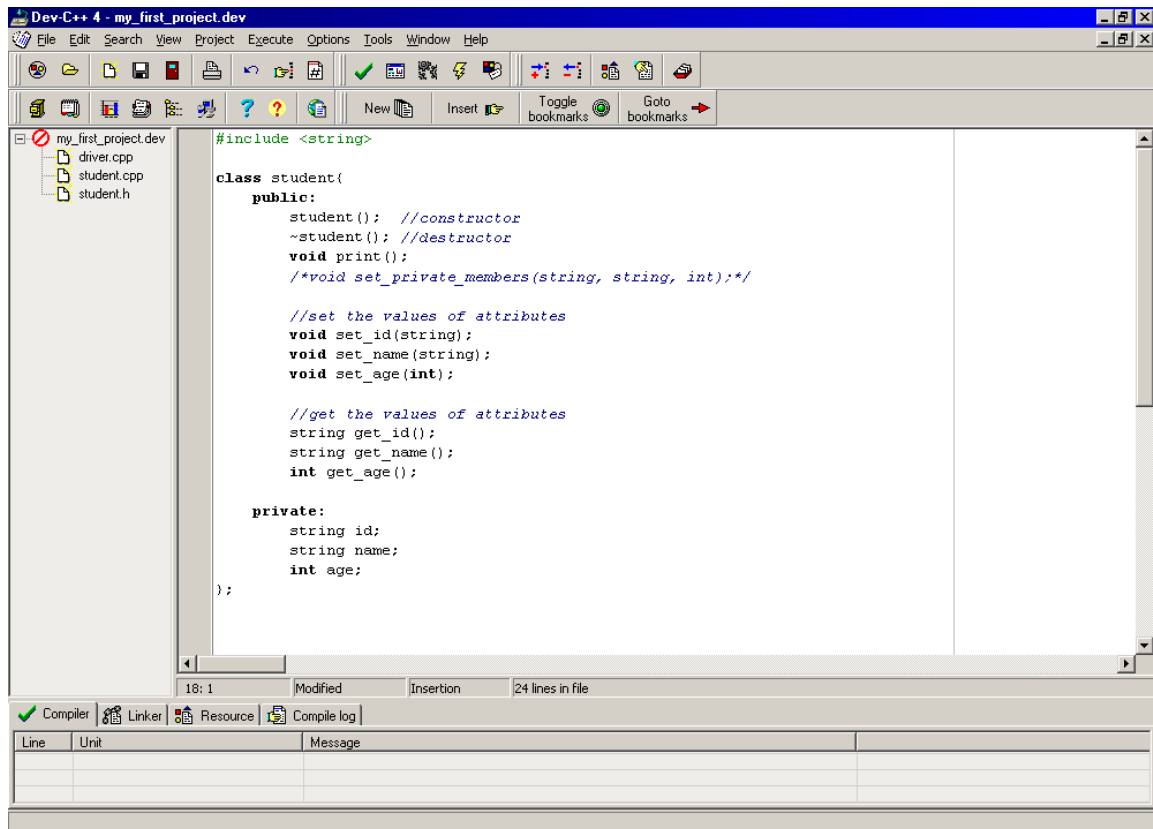
Now right-click your project in the left pane and select *New unit in project* to add new file in the project.



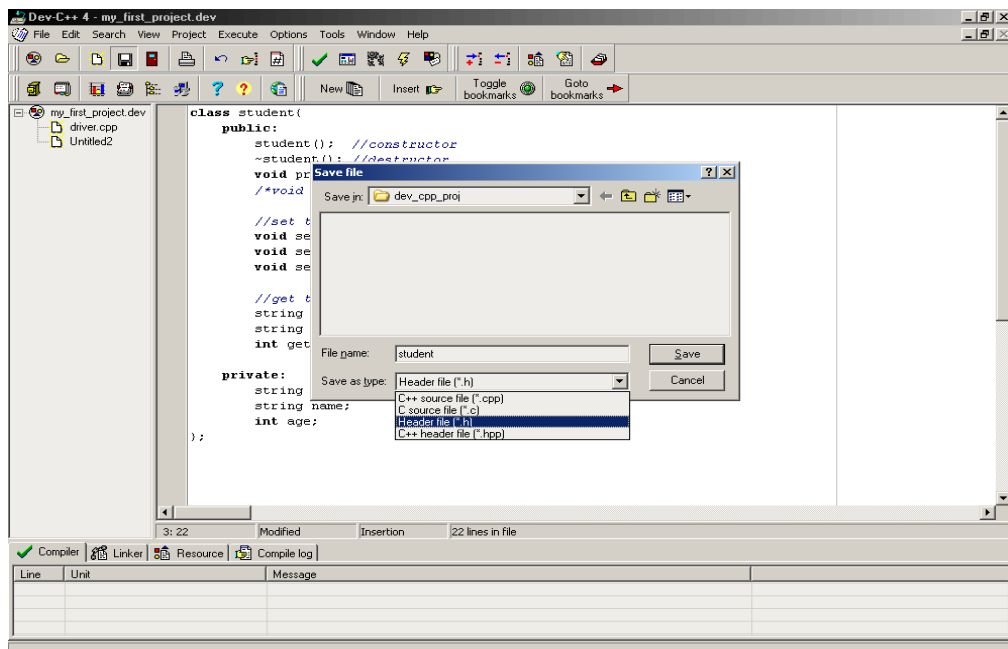
Copy the definition of student class from class_ex6.cpp given in *week 4: class exercises and sample programs* section and paste in newly created file.



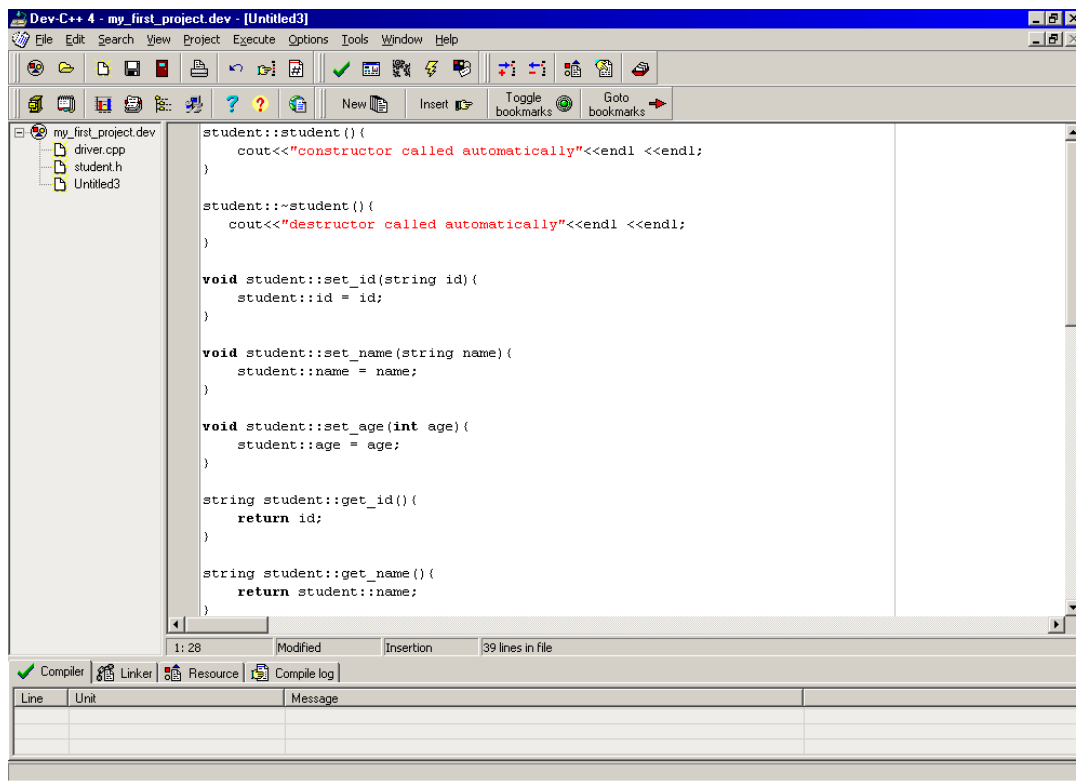
Assume if you just have the following program then you will have to include <string> header file for compilation (why??).



Save this newly created file with the name student. Make sure you change the file type to header file (*.h) as shown in following figure. Class definitions are always saved in header files.

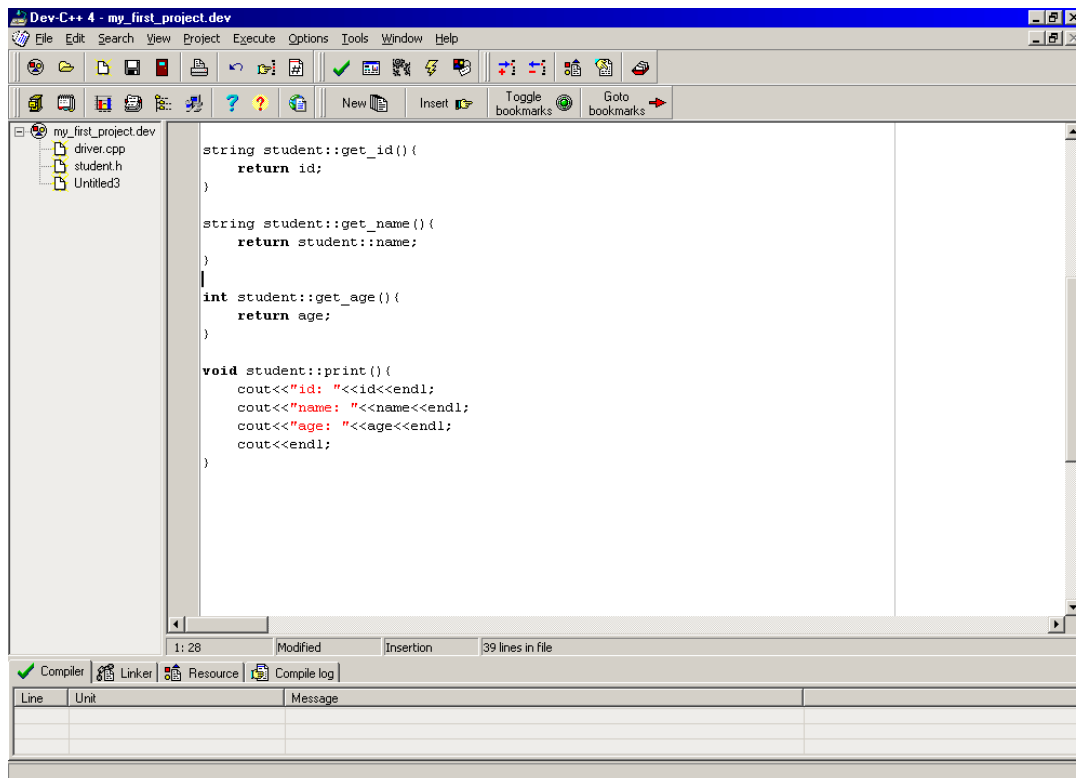


Create one more new file to save the implementation of all methods. Copy the implementation section from the same file class_ex6.cpp. Your pasted code should match the code given in following two figures.



The screenshot shows the Dev-C++ 4 IDE with a project named "my_first_project.dev". The file explorer on the left shows "driver.cpp", "student.h", and "Untitled3". The main editor window displays the implementation of the Student class in "Untitled3". The code includes the constructor, destructor, and three setter methods (set_id, set_name, set_age). The status bar at the bottom indicates "1:28", "Modified", "Insertion", and "39 lines in file".

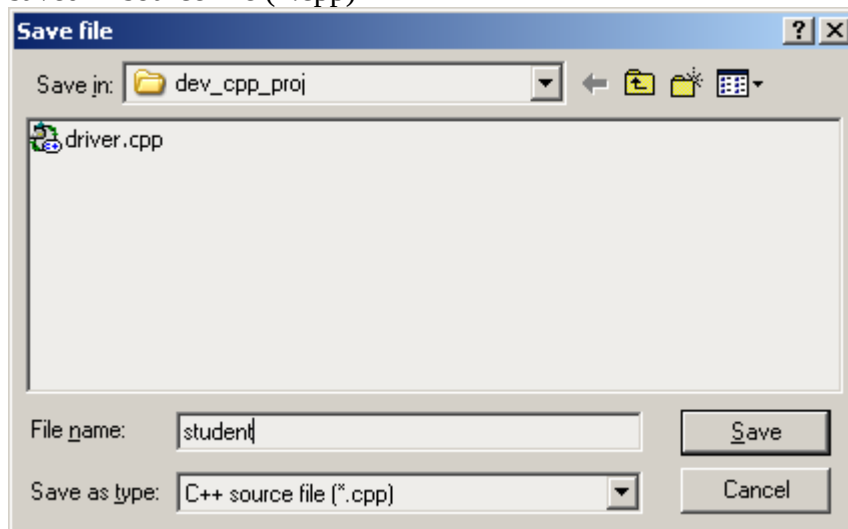
```
student::student() {  
    cout<<"constructor called automatically"<<endl <<endl;  
}  
  
student::~student() {  
    cout<<"destructor called automatically"<<endl <<endl;  
}  
  
void student::set_id(string id) {  
    student::id = id;  
}  
  
void student::set_name(string name) {  
    student::name = name;  
}  
  
void student::set_age(int age) {  
    student::age = age;  
}  
  
string student::get_id() {  
    return id;  
}  
  
string student::get_name() {  
    return student::name;  
}
```



The screenshot shows the Dev-C++ 4 IDE with the same project and file explorer. The main editor window displays the implementation of the Student class in "Untitled3". The code includes the getter methods (get_id, get_name, get_age) and a print method. The status bar at the bottom indicates "1:28", "Modified", "Insertion", and "39 lines in file".

```
string student::get_id() {  
    return id;  
}  
  
string student::get_name() {  
    return student::name;  
}  
  
int student::get_age() {  
    return age;  
}  
  
void student::print() {  
    cout<<"id: "<<id<<endl;  
    cout<<"name: "<<name<<endl;  
    cout<<"age: "<<age<<endl;  
    cout<<endl;  
}
```

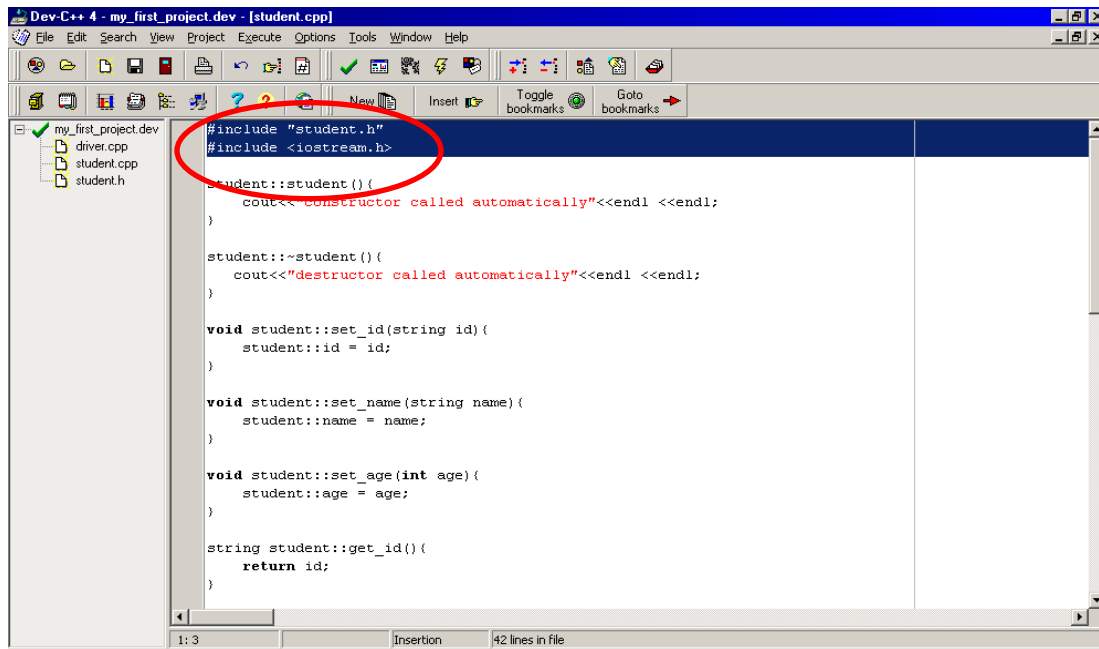
Save this file with the same name of its class definition file. Note that the extension of this file should remain .cpp. Usually, class definition and implementation files have same name, however class definition is saved in header file (*.h) while class implementation is saved in source file (*.cpp)



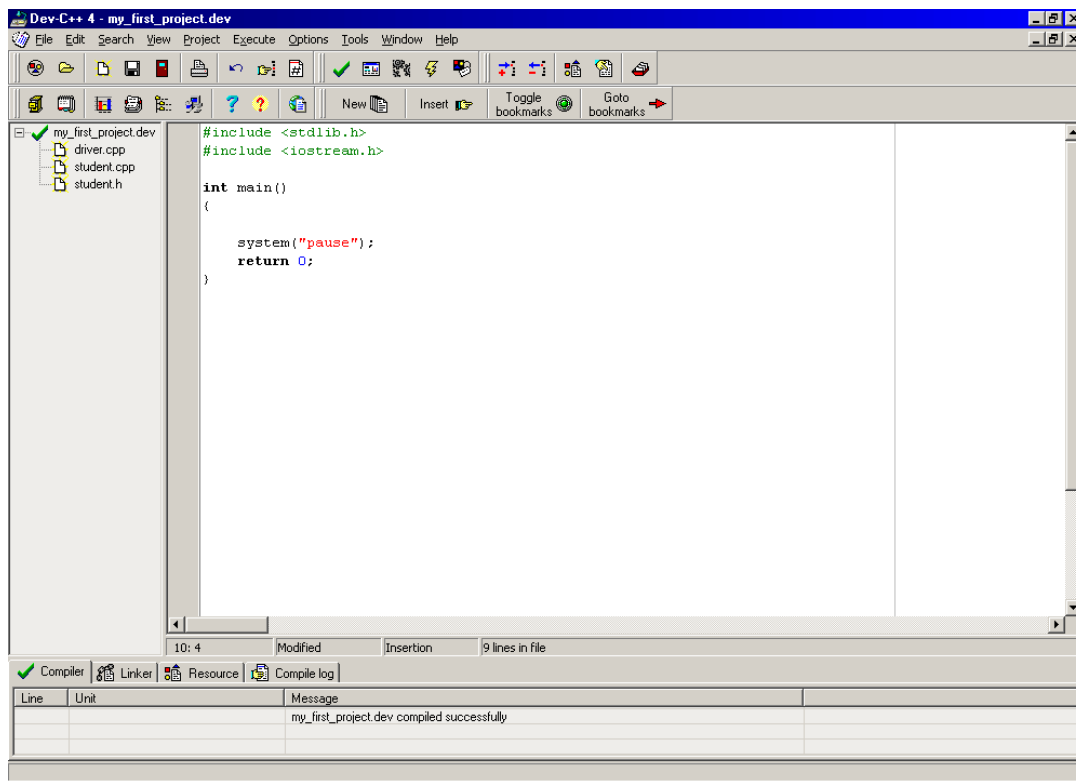
The class implementation file also requires some header files. Can you guess?

Since this file requires definition of student class which is stored in student.h hence student.h file is included. The code in this file also use input and output statements so include <iostream.h>.

Note that user defined header files are enclosed in double quotes rather than angle brackets.



The last step is to write your driver or main program. Since the main doesn't look like the one which you are used to with so lets delete everything from this file and write the general template of main program from scratch.

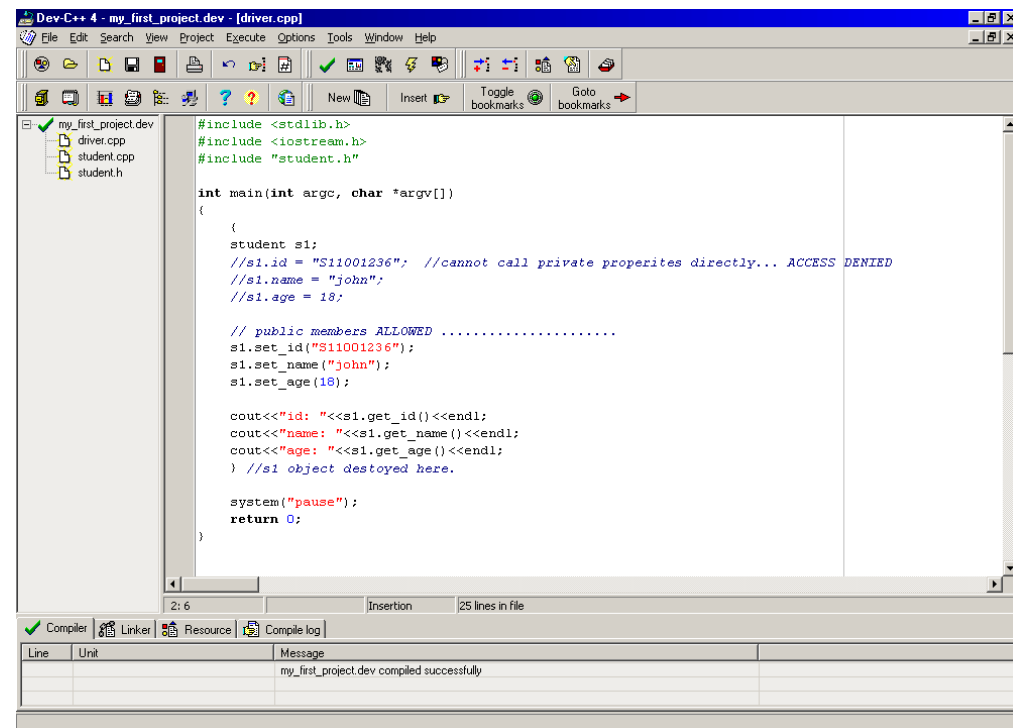


The screenshot shows the Dev-C++ 4 IDE with a project named 'my_first_project.dev'. The file explorer on the left shows 'driver.cpp', 'student.cpp', and 'student.h'. The main editor window displays the content of 'student.h', which contains a simple `main` function that includes `<stdlib.h>` and `<iostream.h>`, calls `system("pause");`, and returns 0. The status bar at the bottom indicates '10: 4 Modified Insertion 9 lines in file'. The compiler output window at the bottom shows a successful compilation message: 'my_first_project.dev compiled successfully'.

```
#include <stdlib.h>
#include <iostream.h>

int main()
{
    system("pause");
    return 0;
}
```

Now copy the content of main from class_ex6.cpp and paste into driver.cpp.



The screenshot shows the Dev-C++ 4 IDE with the same project. The file explorer on the left shows 'driver.cpp', 'student.cpp', and 'student.h'. The main editor window displays the content of 'driver.cpp', which contains a more complex `main` function. It includes `<stdlib.h>`, `<iostream.h>`, and `"student.h"`. The function creates a `student` object `s1`, sets its ID, name, and age, and then prints these values using `cout`. It also includes comments about private properties and public members. The status bar at the bottom indicates '2: 6 Insertion 25 lines in file'. The compiler output window at the bottom shows a successful compilation message: 'my_first_project.dev compiled successfully'.

```
#include <stdlib.h>
#include <iostream.h>
#include "student.h"

int main(int argc, char *argv[])
{
    student s1;
    //s1.id = "S11001236"; //cannot call private properties directly... ACCESS DENIED
    //s1.name = "john";
    //s1.age = 18;

    // public members ALLOWED .....
    s1.set_id("S11001236");
    s1.set_name("john");
    s1.set_age(18);

    cout<<"id: "<<s1.get_id()<<endl;
    cout<<"name: "<<s1.get_name()<<endl;
    cout<<"age: "<<s1.get_age()<<endl;
    } //s1 object destroyed here.

    system("pause");
    return 0;
}
```

The content of class_ex6.cpp is distributed into 3 different files. Now your program is modular and easily manageable.

It is time to compile and run your program. Cheers!

